

Thermal Imagery for Rover Soil Assessment Using a Multipurpose Environmental Chamber Under Simulated Mars Conditions

Raúl Castilla-Arquillo¹, Graduate Student Member, IEEE, Anthony Mandow², Member, IEEE, Carlos J. Pérez-del-Pulgar¹, Member, IEEE, César Álvarez-Llamas¹, José M. Vadillo¹, and Javier Laserna¹

Abstract—Planetary rover missions on Mars have suffered entrapments and serious mobility incidents due to soil assessment limitations of stereo RGB cameras, which cannot characterize relevant physical phenomena such as thermal behavior that depend on granularity and cohesion. In particular, thermal inertia estimations are already being used to assess geophysical properties from 1-D low-resolution measurements by onboard thermopiles. However, no high-resolution measurements are currently available to characterize Martian soils for safer navigation in future missions, so new experimental methods are required to capture and analyze thermal images with planetary conditions in Earth-based experiments. In this work, we propose a novel measurement system configuration and experimental methodology to capture thermal images using isolated multipurpose environmental chambers (MECs) to replicate the temperature and pressure conditions of Mars. Furthermore, the system has allowed to measure diurnal cycles for four soil types of known physical characteristics under Martian and Earth pressures to perform a unique quantitative analysis and comparison of thermal behavior and thermal inertia for soil assessment. Even if no actual Martian infrared (IR) images are available for comparison, results indicate a correlation between granularity and thermal inertia that is consistent with available thermopile measurements recorded by rover's onsite. Furthermore, the set of measurements acquired in the experiments has been made available to the scientific community.

Index Terms—Measurement equipment, multipurpose environmental chamber (MEC), planetary robotics, thermal imagery, thermal inertia.

I. INTRODUCTION

REMOTE assessment of soil characteristics can be crucial for the safety and efficiency of a broad range of tasks related to the navigation of planetary mobile robots such as odometry, environment mapping, or energetic consumption [1]. Soil assessment is essential to mitigate the risks

of slippage, skidding, and entrapment on granular terrains, which caused delays and considerable mobility issues during the Curiosity and Spirit rover missions [2], [3], [4]. For instance, the Spirit rover became entrapped while trying to traverse undetected loose sand hidden under a thin surface layer of duricrust and subsequently stopped responding to commands [5], highlighting the importance of accurate soil assessment for future missions.

Remote sensors onboard mobile robots, such as RGB stereo cameras or 3-D laser scanners [6], can be used to infer soil characteristics like roughness and slope [7], [8] without soil contact, but do not allow assessment of relevant subsurface properties for traversability (e.g., granularity and soil cohesion). In this sense, infrared (IR) remote sensors such as thermopiles and thermal cameras have provided relevant data to infer subsurface properties, as the thermal behavior of the surface is influenced by the composition of deeper soil layers [9]. Thus, soil composition can be remotely inferred by estimating the soil thermal inertia, which is the key material property that affects the surface thermal behavior, over daily cycles of temperature variations [10]. Thermopiles are being actively used in the Curiosity and Perseverance rovers to perform measurements of Martian surface thermal behavior to assess the geophysical properties of areas of interest. Remote temperature measurements taken during their traverses have been employed to infer the thermal inertia of several types of soils not only for scientific purposes [11], [12] but also for rover slip prediction [13]. However, thermopiles measure 1-D data from surfaces in the order of several square meters, which poses limitations for characterizing heterogeneous soils.

Furthermore, quantitative analysis of soil characteristics inferred from thermal inertia is very challenging in a planetary context because the actual terrain is not accessible for experimental validation [13]. Capturing high-resolution thermal images of soils of known physical characteristics under simulated planetary conditions is important for experiments aimed at estimating thermal inertia, which is notably influenced by atmospheric pressure. To the best of our knowledge, no comparative analysis of diurnal cycle surface thermal behavior has been performed for different soils under Martian and Earth pressures.

In this context, multipurpose environmental chambers (MECs) are experimental measurement equipment that can function under conditions that closely resemble the tempera-

Manuscript received 24 July 2023; revised 15 November 2023; accepted 30 November 2023. Date of publication 25 December 2023; date of current version 15 January 2024. This work was supported in part by the Andalusian Regional Government through the Project Intelligent Multimodal Sensor for Identification of Terramechanic Characteristics in Off-Road Vehicles (IMSITER) under Grant P18-RT-991 and in part by the Open Access funding provided by Universidad de Málaga/CBUA. The Associate Editor coordinating the review process was Dr. Libing Bai. (Corresponding author: Carlos J. Pérez-del-Pulgar.)

Raúl Castilla-Arquillo, Anthony Mandow, and Carlos J. Pérez-del-Pulgar are with the Institute for Mechatronics Engineering and Cyber-Physical Systems (IMECH.UMA), University of Málaga, Andalucía Tech, 29070 Málaga, Spain.

César Álvarez-Llamas, José M. Vadillo, and Javier Laserna are with the UMALASERLAB, Department of Analytical Chemistry, University of Málaga, 29010 Málaga, Spain (e-mail: carlosperez@uma.es).

Digital Object Identifier 10.1109/TIM.2023.3346528



Fig. 1. UMALASERLAB Mars Environment Chamber used in the experiments.

tures and pressures found on other planets, such as Mars [14], [15], [16], but reported experiments are limited to constant conditions. These chambers are constructed with isolated walls so that the effects of external conditions such as temperature, pressure, or humidity are negligible, but novel instrument setup methods are required to perform remote thermal measurements through an appropriate IR viewport to withstand the pressure differential and temperatures reached by the MEC and to simulate diurnal temperature cycles.

This article proposes a measurement system configuration and experimental methodology to capture thermal images using MECs to replicate the temperature and pressure conditions of Martian diurnal cycles. The instrument setup and experimental methodology have been applied to the UMALASERLAB Multipurpose Thermal Vacuum Chamber [17] (see Fig. 1), which can also replicate the atmospheric CO_2 composition of Mars. Moreover, we apply the proposed system to offer a novel quantitative analysis of thermal behavior and thermal inertia of four types of soil with known physical characteristics for diurnal cycles with Martian and Earth pressures. Furthermore, the dataset resulting from the experiments has been made available to the scientific community.

This article is organized as follows. Section II discusses related work. Section III reviews thermal inertia formulation. Section IV presents the proposed MEC-based remote thermal measurement system. Section V describes the experimental setup. Section VI presents an experimental analysis. Finally, Section VII offers conclusions and provides insight into future works.

II. RELATED WORK

Planetary exploration rovers are fit with remote sensors such as RGB stereo cameras [18] for mission planning assistance, but these cameras have not been suitable to avoid the risks of slippage, skidding, and entrapment on granular terrains, which have resulted in delays and considerable mobility issues during the Curiosity and Spirit rover missions [2], [3], [4]. As pointed out in [13], the Spirit rover could have avoided

getting trapped in the sand if a thermal camera had been used to assess the duricrust-covered sandy area by contrasting thermal inertia because, unlike visual surface appearance, this physical phenomenon is directly correlated to properties such as particle size, soil cohesion, porosity, and bedrock abundance, which are essential for vehicle slippage prevention and safe mobility.

Thermal inertia represents a combination of physical properties that are not directly measurable in practice, so simplified estimations based on surface temperature observations (either from satellite measurements or on-site data) are required. Thermal inertia can be estimated by fitting the measured thermal daily cycles with the temperature obtained from a surface thermal model, which entails solving numerical models by iterating the values of several unknown parameters until the model's curve fits the measurements [19]. Alternatively, the apparent thermal inertia (ATI) [20] allows a simpler computation, but it does not consider the surface net heat flux. In this work, we use a method based on the daily amplitude of surface net heat flux and temperature to calculate the thermal inertia of a surface subjected to the Sun's heating [21].

Satellite-based measurements of thermal inertia have provided a means to determine the physical properties of the Earth's surface on a global scale, which would be impractical to obtain through on-site measurements [20], [22]. Similarly, satellite measurements of the Martian surface have been helpful to identify potential areas of scientific interest for future exploration [23], [24]. Satellite measurements, although valuable for large-scale assessments, often lack the required resolution for robotic navigation and can be influenced by clouds and other atmospheric conditions.

As for on-site measurements, the deployment of thermopiles on Mars rovers has been used to identify soil characteristics of traversed paths at higher resolution than satellites by acquiring diurnal cycle measurements [25], [26]. The Curiosity rover uses the ground temperature sensor (GTS), positioned at a height of 1.6 m, with a downward-looking field of view (FOV) covering 100 m^2 of terrain [27]. Furthermore, the Perseverance rover features the thermal IR sensor (TIRS), a combination of five thermopiles, one of which measures ground temperature from a height of 1.5 m with an FOV of approximately 3 m^2 [28]. Nonetheless, 1-D IR sensors such as thermopiles offer insufficient resolution to distinguish between adjacent heterogeneous soil regions [13].

Current rover missions already incorporate multispectral 2-D cameras for narrowband visible/near-IR (VNIR) for mineralogical measurements [18], [29], but long-wave IR (LWIR) imagery is needed for thermal behavior assessment. On Earth, on-site thermal cameras have been employed to estimate diurnal thermal inertia of different soils, even if results were limited by the effect of Earth pressure [30], [31].

Further development on planetary soil assessment with LWIR cameras is hindered by the lack of actual thermal images from Mars rovers [32]. This is especially critical in view of the rapid development of machine learning methods, which demand large amounts of training measurements [33], [34], [35]. This problem has been addressed by creating synthetic Martian data using generative neural networks [36].

Particularly, synthetic IR Martian images for thermal inertia estimation were generated from a multimodal combination of Earth data, including RGB imagery [32]. Alternatively, more accurate planetary conditions such as pressure and atmospheric composition can be physically simulated in MECs. For instance, JPL developed a large facility for the simulation of intense solar radiation in interplanetary vacuum conditions [14]. More recently, a small-scale MEC was developed to simulate planetary pressure and temperature conditions for testing instrumentation, procedures, and materials, called SpaceQ [15]. Moreover, the Mars MECs in [16] and [17] are used for VNIR and laser-induced breakdown spectroscopy (LIBS) measurements. Since this equipment was not conceived neither for LWIR cameras nor to simulate diurnal cycles, novel system configurations, and experimental methodologies are needed to adapt MECs for thermal behavior measurements under planetary conditions.

The major novelties and advances of this work in the instrumentation and measurement context are the following.

- 1) We offer an experimental methodology for MECs to simulate Martian diurnal cycles with an appropriate IR viewport setup for remote thermal measurements.
- 2) We provide technical details on the implementation of the experimental methodology in the UMALASERLAB MEC [17].
- 3) We provide a diurnal cycle quantitative soil analysis from high-resolution thermal images captured under Martian and Earth conditions (i.e., pressure and atmospheric composition).
- 4) We offer a public dataset of the radiometric images and data resulting from the MEC experiments.

To the best knowledge of the authors, no previous works have addressed these issues.

III. THERMAL INERTIA

This section reviews thermal inertia concepts, the use of the thermal diffusion equation to model the Martian surface thermal behavior, and a method to estimate thermal inertia based on surface temperature gradients.

A. Definition and Pressure Dependence

Thermal inertia, I , is defined as follows:

$$I = \sqrt{k\rho c} \quad (1)$$

where k is the bulk thermal conductivity, ρ is the bulk density, and c is the soil specific heat capacity. Thermal inertia is the property of a material that affects the resistance of a soil to change its temperature. A higher thermal inertia value means a slower heating of the soil. Thermal conductivity is the parameter that mainly influences thermal inertia, which is affected by three different heat transfer mechanisms [10]

$$k = k_r + k_c + k_g \quad (2)$$

where k_r is the transfer across pore spaces, k_c is the conduction between grains contact areas, and k_g is the conduction of the gas which fills the pores between grains. Pressure greatly determines which term acquires the

most relevance. Gas conduction (k_g) dominates at pressures between 0.1 and 1000 mbar, where there is a near-linear relationship between particle size and thermal conductivity for granular soils [37], [38]. In this case, loose granular soils have lower thermal inertia than compacted rocky soils [39]. However, the relationship is not so strong at pressures higher than 1000 mbar. Thus, it is much easier to estimate the soil characteristics based on thermal inertia at Martian pressure than at Earth pressure. On the other hand, it would be significantly harder on the Moon, where the conduction mechanism is not presented due to vacuum.

B. Martian Surface Behavior

Martian surface thermal behavior can be expressed as a boundary condition on the thermal diffusion equation derived from its surface energy budget

$$\begin{aligned} G &= -I \sqrt{\frac{\pi}{P}} \frac{\partial T}{\partial Z'} \Big|_{Z'=0} \\ &= (1 - A)R_{sw} - \epsilon\sigma_B T_s^4 + \epsilon R_{lw} - F_{CO_2} \end{aligned} \quad (3)$$

where G is the net heat flux expressed in W/m^2 , A is the albedo, σ_B is the Stefan-Boltzmann's constant, R_{sw} is the downwelling shortwave (SW) radiation absorbed from the Sun, R_{lw} is the downwelling longwave (LW) radiation emitted by the atmosphere and the Sun, ϵ is the thermal emissivity, F_{CO_2} is the seasonal CO_2 condensation, P is the period of a diurnal cycle, T_s is the surface temperature, and the term $(\partial T/\partial Z')|_{Z'=0}$ is the temperature gradient evaluated at the surface of the terrain, being Z' the distance into the terrain normalized to the thermal skin depth. The sign convention is to use a positive sign when modeling the heating of the terrain and a negative sign when modeling its cooling.

The F_{CO_2} term of (3) is negligible for Martian surfaces between equatorial and mid-latitudes that present no frost. Moreover, the downwelling LW radiation is not considered, as previous on-site measurements have shown it to be an order of magnitude smaller than the rest of the terms [11]. Thus, the simplified equation for the soil thermal behavior is

$$G = -I \sqrt{\frac{\pi}{P}} \frac{\partial T}{\partial Z'} \Big|_{Z'=0} = (1 - A)R_{sw} - \epsilon\sigma_B T_s^4 \quad (4)$$

where the soil thermal behavior depends on the SW incident Sun's radiation, thermal inertia, and the surface radiative emission.

C. Thermal Inertia Estimation

In this work, we estimate thermal inertia based on the daily amplitude of surface net heat flux and temperature of a surface subjected to the Sun's heating [21]. This method considers a sinusoidal approximation of the Earth's net heat flux and surface temperatures for a diurnal period P , which also applies to Mars' heat fluxes and temperatures [11]. The thermal inertia of a given soil is estimated as

$$I_{sin} = \frac{\Delta G_s}{\Delta T_s \sqrt{2\pi/P}} \quad (5)$$

where $\Delta T_s = T_{max} - T_{min}$ and the net heat flux is expressed as $\Delta G_s = G_{max} - G_{min}$, being G_{max} and G_{min} the maximum and

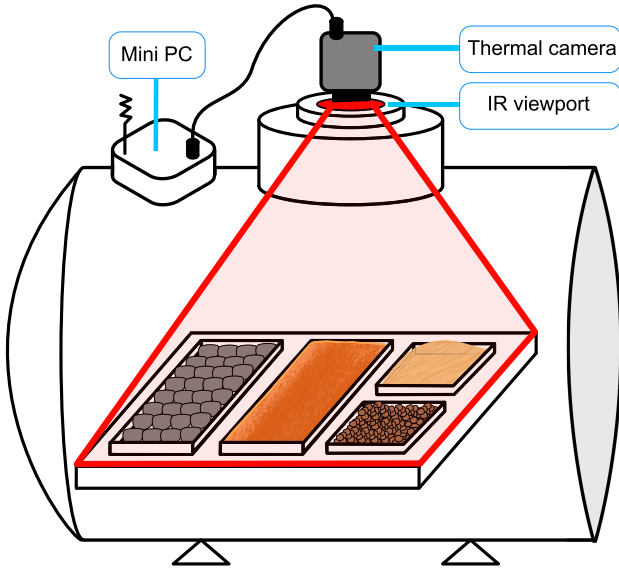


Fig. 2. Schematic of the proposed physical MEC-based configuration.

minimum values of the net heat flux, respectively. The result can be multiplied by a 4186 coefficient to express the inertia in thermal inertia units, $\text{tiu} \equiv (W s^{1/2}/m^2 K)$. Throughout this work, I_{sin} will always be expressed in tiu.

IV. MEC-BASED REMOTE THERMAL MEASUREMENT SYSTEM

In this section, we propose an MEC-based high-resolution remote thermal measurement system to physically simulate soil thermal behavior over diurnal cycles under planetary conditions of pressure and atmospheric composition. The measurement system consists of a physical MEC-based configuration (see Fig. 2) and an experimental methodology. The algorithmic representation of the proposed methodology is shown in Fig. 3.

A. Physical MEC-Based Configuration

The physical MEC-based configuration (see Fig. 2) allows to perform high-resolution remote temperature measurements under the extreme conditions produced by the MEC. This configuration consists of an inner plate where the sample bins are placed, an external thermal camera connected to a Mini PC for data collection, and an IR viewport. The viewport must allow the IR range from 8 to 14 μm to pass through with minimal losses. Furthermore, it must withstand the pressure differential and temperatures reached by the MEC.

B. Experimental Methodology

The experimental methodology (see Fig. 3) consists of three major sequential tasks: the study of the physical characteristics of the soils and preparation of the MEC setup, the adjustment of the inner pressure according to the environment to be replicated, and the physical simulation within the MEC, followed by the estimation of the thermal inertias based on the resulting measurements.

During the MEC preparation task, soils with previously studied physical properties (e.g., granularity) are selected and

deposited into the sample bins. Subsequently, these sample bins are positioned on the plate. At this point, thermal insulation between the sample bins and the plate must be guaranteed to prevent potential IR reflections that could distort the measurements. The thermal camera is mounted on the viewport, and its housing is grounded to prevent electrostatic charges generated by the MEC pumps. The thermal camera is then calibrated to ensure accurate measurements of the sample bin surfaces. Finally, the MEC is sealed.

In the pressure adjustment task, procedures vary depending on the pressure and temperature range for the experiments. In the case of Earth pressure experiments with temperatures exceeding 0 $^{\circ}\text{C}$, the physical simulation can start. However, if different conditions are required, such as Mars pressure or a specific air composition (i.e., 95% of CO_2 for Mars), the air must be evacuated to create a vacuum environment, followed by the introduction of moisture-free air with the desired composition. This method safeguards the MEC internal systems from potential issues related to the freezing of air moisture.

In the physical simulation task, temperatures are set for the MEC heaters to achieve sinusoidal soil sample temperatures resembling those encountered in a real diurnal cycle. The surface energy budget for each sample bin within the MEC can be described as a function of the radiative flux generated by the MEC heaters, as shown in the following equation:

$$G = -I \sqrt{\frac{\pi}{P}} \left. \frac{\partial T}{\partial Z'} \right|_{Z'=0} = \epsilon \sigma_B T_{\text{heater}}^4 - \epsilon \sigma_B T_{s_i}^4 \quad (6)$$

where T_{s_i} represents the mean surface temperature of each soil sample and T_{heater} corresponds to the temperature of the MEC heaters. Given that the MEC is a closed space with no air movement, natural convection is considered negligible. Within the MEC, the radiation term $\epsilon \sigma_B T_{\text{heater}}^4$ simulates the active Sun heating, which corresponds to the term $(1 - A)R_{\text{sw}}$ of (4).

Once the simulated diurnal cycle is complete, the thermal inertia of each soil within the sample bins can be estimated by calculating corresponding the surface temperature difference, denoted as ΔT_s , and the difference in the net heat flux, ΔG_s , using (6). Next, these terms can be applied to (5) to determine the thermal inertia, I_{sin} , in tiu.

V. EXPERIMENTAL SETUP

The measurement system proposed in Section IV was employed to replicate the thermal behavior of four soil samples with known physical properties in a diurnal cycle, under conditions simulating both Earth and Martian environments. This section presents the integration of hardware components to evaluate the proposed system as well as the selection of soil samples.

A. Equipment

The UMALASERLAB MEC (see Fig. 1) is a stainless-steel cylinder of 12 m of length and 1.6 m of diameter and viewports on the top and sides [17]. It is equipped with an inner spot-gridded thermal jacket or heater which contains a cooling fluid that can reach a temperature in the range of -72°C to 127°C , at a rate of $1^{\circ}\text{C}/\text{min}$. The air inside can be pumped

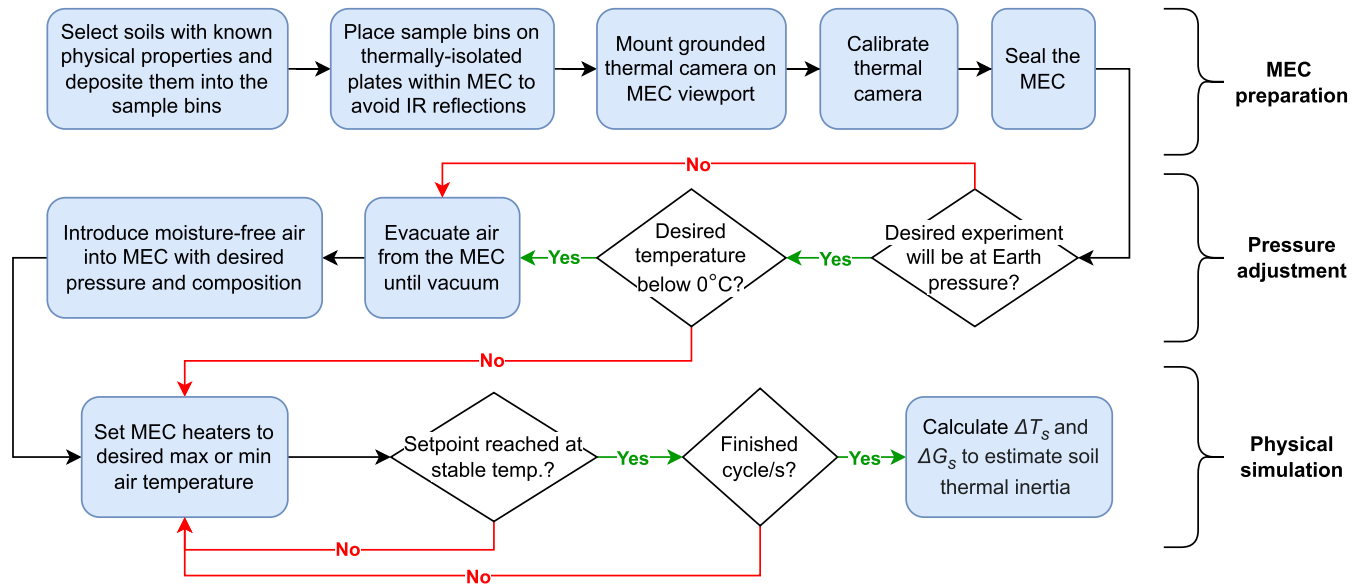


Fig. 3. Flowchart of the proposed experimental methodology.

out of the chamber until a pressure of 10^{-4} mbar is reached and can be replaced by CO_2 to simulate the composition of the Mars atmosphere. The outer wall of the chamber has a sealed vacuum compartment to isolate the interior from external environmental conditions. Therefore, the effect of other conditions such as external temperature, pressure, or humidity is negligible. The MEC has ISO160 K-compliant glass viewports for VNIR measurements. Moreover, vacuum-compliant thermocouple gauges are set in the center of its core to measure the air temperature. Additionally, the MEC has a stainless steel plate on rails that allows a payload of up to 70 kg.

The thermal vision camera is a PI-640i by Optris based on uncooled microbolometer technology. It is a 320-g LWIR camera that works in the spectral range of 8–14 μm , has a resolution of 640×480 pixels, and a germanium optic with an FOV of $60^\circ \times 45^\circ$. It can measure temperatures from -20°C to 900°C with a thermal sensitivity of 0.04°C . We selected this camera due to its high resolution and lightweight, making it suitable for mobile and aerial robots. However, this uncooled camera does not provide temperature measurements below -20°C , which limited the absolute minimum temperature to which soil samples could be subjected.

The thermal camera was connected to a Mini PC Intel NUC with an Intel Core i5 processor of 1.8 GHz and 16 GB of RAM running the software Optris PIX connect. We adjusted the focal length of its optic by using a warm body (i.e., a hand) placed on the plate as a reference. The thermal camera geometric and radiometric calibrations were provided by the manufacturer. The sample bins were placed on the plate perpendicular to the thermal camera at a distance of around 1.3 m to have an undistorted view of their surfaces. The plate surface was covered with insulating cardboard and a thick black fabric to avoid IR reflections from the steel.

We designed and developed an LWIR viewport adapter to replace one of the ISO160 K viewports (see Fig. 4) to make measurements from outside the MEC using the

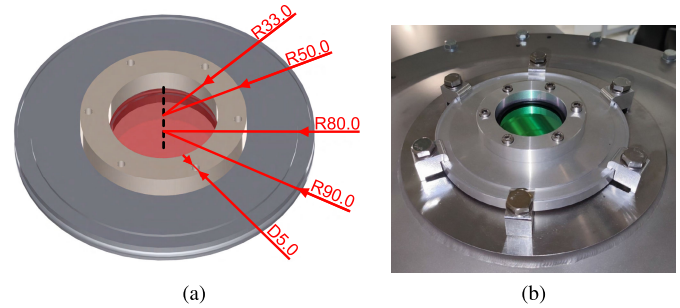


Fig. 4. Customized viewport adapter. (a) 3-D model design (dimensions in millimeters): toroid frame (dark gray), germanium window (red), and clamping ring (light gray). (b) Image of the clamped adapter.

thermal camera. This IR window adapter keeps the inside of the MEC sealed while letting the LWIR range from 8- to 14- μm radiation pass through. We chose an anti-reflection coated germanium circular optic model GEW16AR.20 by MKS Instruments due to its high mechanical resistance and its ability to withstand abrupt thermal changes. We selected a diameter of 74.9 and 5.0 mm of thickness in order to comply with the minimum thickness required to avoid reaching the germanium's fracture strength caused by the pressure differential between the environment and Martian pressure inside the MEC [40]. Furthermore, an aluminum toroid frame with a screwed clamping ring was crafted to fix the germanium window into the MEC upper ISO160 K-compliant viewports.

B. Soil Samples

Four sample bins with soils of different characteristics were selected for the experiments (see Fig. 5). Table I summarizes characteristics, sorted by granularity. Three of the bins contained granular soils and one contained an example of bedrock. We plotted a granularity chart of the granular soils (see Fig. 6) by passing them through sieves with grids of different sizes. In terms of homogeneity, soil C is the most homogeneous,

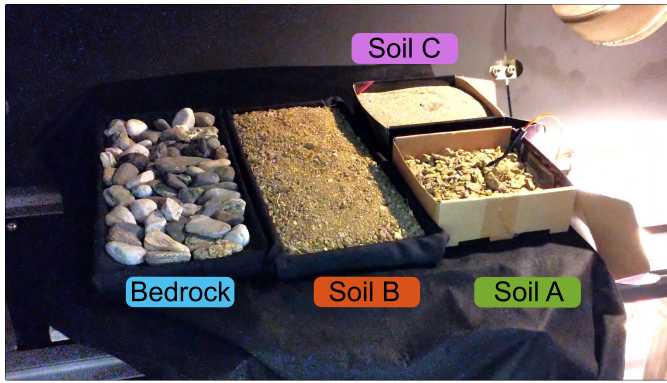


Fig. 5. Sample bins of soils of different granularities introduced into the MEC.

TABLE I
SAMPLE BINS CHARACTERISTICS

Sample	Granularity (mm)	Density (g/ml)	Bin size (cm)
Bedrock	40.0 - 50.0	2.94	53 x 23 x 3.5
Soil A	3.0 - 5.0	1.43	22 x 22 x 7
Soil B	1.3 - 2.0	1.40	55 x 23 x 3.5
Soil C	0.7 - 1.0	1.71	22 x 22 x 7

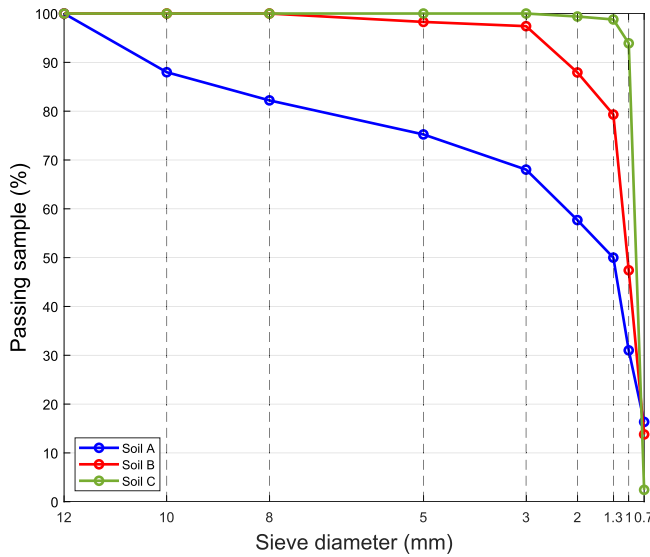


Fig. 6. Granularity chart of the soils.

as more than 90% of its grains have a diameter of 0.7–1 mm. It is followed by soil B, whose grains are mostly concentrated on the size of less than 2.0 mm. Finally, soil A is classified as the most heterogeneous, as it consists of a mixture of several grain sizes.

VI. EXPERIMENTS

In this section, we apply the proposed system configuration and methodology to provide a diurnal cycle quantitative soil analysis from high-resolution thermal images captured under Martian and Earth conditions (i.e., pressure and atmospheric composition). In particular, the captured thermal images are processed to analyze the soils thermal behavior and to estimate their thermal inertia. For validation purposes, the thermal

TABLE II
DESCRIPTION OF THE MEC EXPERIMENTS

	Experiment	Press. (mbar)	Subsurf.	P_e (min)
Pair-1	#1 Earth-like	1000	Soil A	296
	#2 Mars-like	8	Soil A	297
Pair-2	#3 Earth-like	1000	Soil C	320
	#4 Mars-like	8	Soil C	360

inertia values acquired in the experiments conducted under Martian conditions are compared with the estimations of thermal inertia derived from the on-site surface temperature measurements taken by the Perseverance rover.

A. Description of the Experiments

Two sets of experiments, namely, pair-1 and pair-2, were conducted within the MEC using the soils specified in Table I to ensure redundancy in measurements. These experiment pairs encompassed conditions representing Earth-standard pressure (1000 mbar) (1 and 3) as well as Mars-like pressure (8 mbar) (2 and 4). Table II provides an overview of the key characteristics of each experiment. Due to limitations in MEC connectivity, subsurface temperature data could only be equatorial for one of the soils on each experiment. For pair-1 and pair-2, the subsurface thermocouple gauge was positioned at a depth of 3 cm in soils A and C, respectively.

During the Mars-like experiments, the MEC was filled with air matching Mars' carbon dioxide (CO_2) atmospheric composition (i.e., 95%). Environmental measurements taken near the equatorial region on Mars by the Mars Science Laboratory (MSL) indicated mean daily air temperatures around -50°C with approximate amplitudes of 60°C [41]. Thus, sinusoidal temperatures of similar amplitude were simulated in diurnal cycles of experimental actuation period P_e by manual input of constant heating and cooling setpoints for MEC actuation.

Soil surface temperatures were measured by means of the Optris PI-640i thermal camera. The measurements were made assuming no prior knowledge of the soils, as would be the case in actual rover soil assessment implementations. Thus, emissivity (ϵ) was considered to be unitary and the albedo (A) to be zero for all the soils, according to Kirchhoff's law: $1 = A + \epsilon$ [42]. Polygonal areas delimiting each soil were defined in the acquired thermal images, with the pixels containing the thermocouple gauge being excluded to avoid affecting temperature measurements.

Figs. 7–10 show the diurnal cycle temperatures for all soil types from Experiments 1–4, respectively. Besides, Figs. 11 and 12 present surface and subsurface temperature readings for soil A (Experiments 1 and 2) and soil C (Experiments 3 and 4), respectively. All the figures display the heater, setpoint, and air temperatures. The transient phase is considered to conclude once an inflection point is reached in the rising heater temperature response.

Table III provides the mean soil surface temperatures for the pixels within the corresponding polygonal area, along with the standard deviations for the four experiments. In this table, $\Delta T_s = T_{\max} - T_{\text{init}}$, where $T_{\max}$ represents the maximum

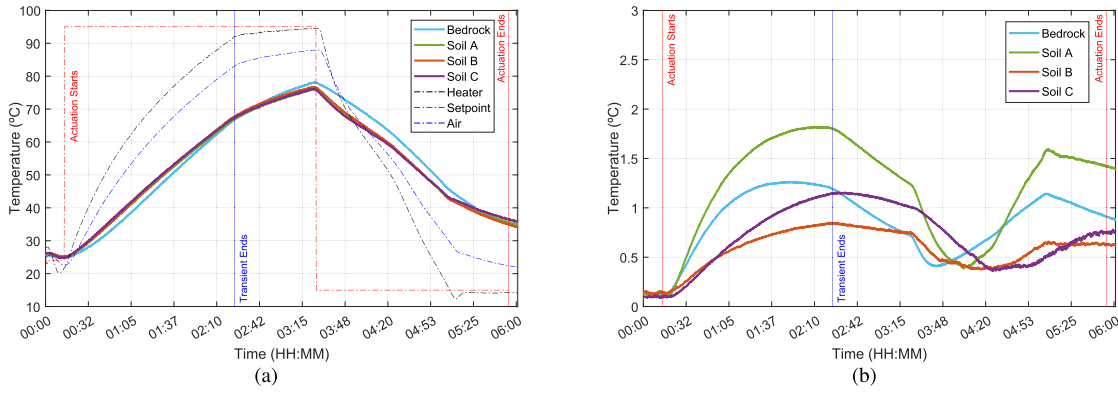


Fig. 7. Diurnal cycle temperatures for Experiment 1 at Earth's pressure ($p = 1000$ mbar). (a) Mean temperature. (b) Standard deviation.

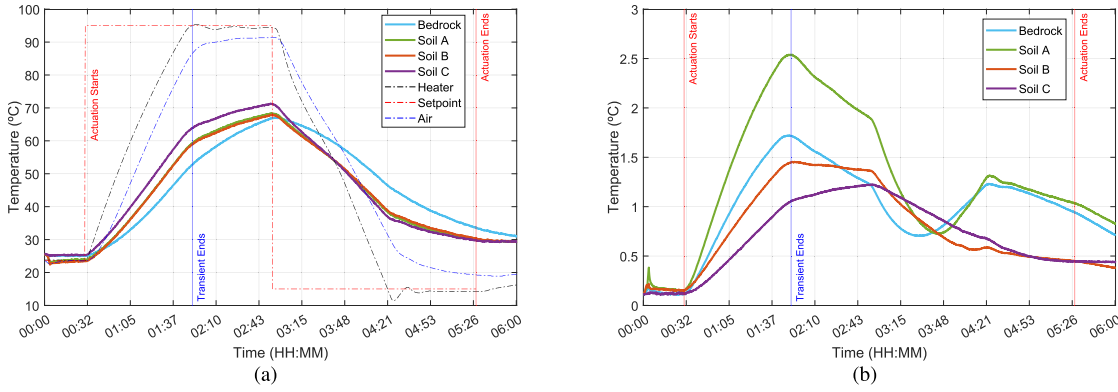


Fig. 8. Diurnal cycle temperatures for Experiment 2 at Martian pressure ($p = 8$ mbar). (a) Mean temperature. (b) Standard deviation.

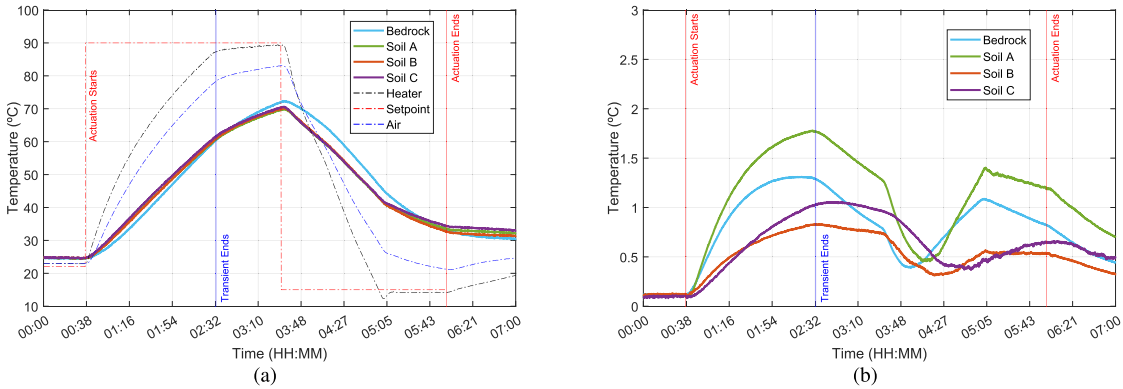


Fig. 9. Diurnal cycle temperatures for Experiment 3 at Earth's pressure ($p = 1000$ mbar). (a) Mean temperature. (b) Standard deviation.

mean temperature and T_{init} is the mean temperature at the beginning of the actuation. T_{tran} denotes the standard deviation temperature for each soil when the actuation transient phase concludes. The net heat fluxes were calculated by applying the surface energy budget equation for each sample bin within the MEC, as described in (6). This computation used the mean temperatures of the soils and the MEC heater temperatures recorded during the experiments. Additionally, the increase in net heat flux during the experiments was determined as $\Delta G_s = G_{\text{max}} - G_{\text{init}}$, where G_{max} represents the maximum net heat flux and G_{init} is the net heat flux at the beginning of the actuation. The thermal inertia estimates in (5), denoted as I_{sin} , were calculated based on the values of ΔT_s , ΔG_s , and P_e obtained during the experiments.

B. Quantitative Analysis of Thermal Behavior and Inertia

This section analyzes thermal behavior measurements in terms of the surface mean temperatures, the soil standard deviation curves, the subsurface temperatures, and thermal inertia.

First, we compare the increment in the surface mean temperature, denoted as ΔT_s , of pair-1 [Figs. 7(a) and 8(a)]. These graphs show that only the bedrock is distinguishable at Earth's pressure, as all the granular soils exhibit similar values. In contrast, at Martian pressure, three prominent groups can be observed; from highest to lowest: soil C; soils A and B; and bedrock. Furthermore, both graphs reveal a slight temporal delay in the bedrock temperature curve compared to the other soils. The same analysis can be applied to the graphs of pair-2 [Figs. 9(a) and 10(a)].

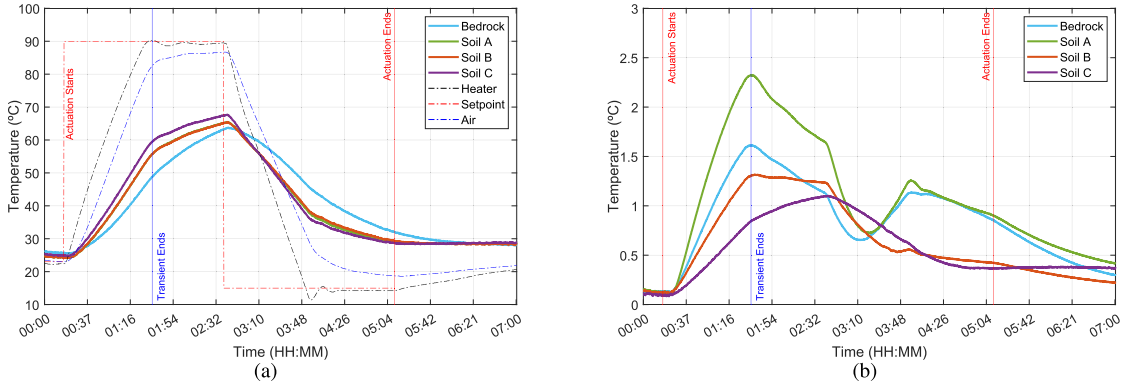


Fig. 10. Diurnal cycle temperatures for Experiment 4 at Martian pressure ($p = 8$ mbar). (a) Mean temperature. (b) Standard deviation.

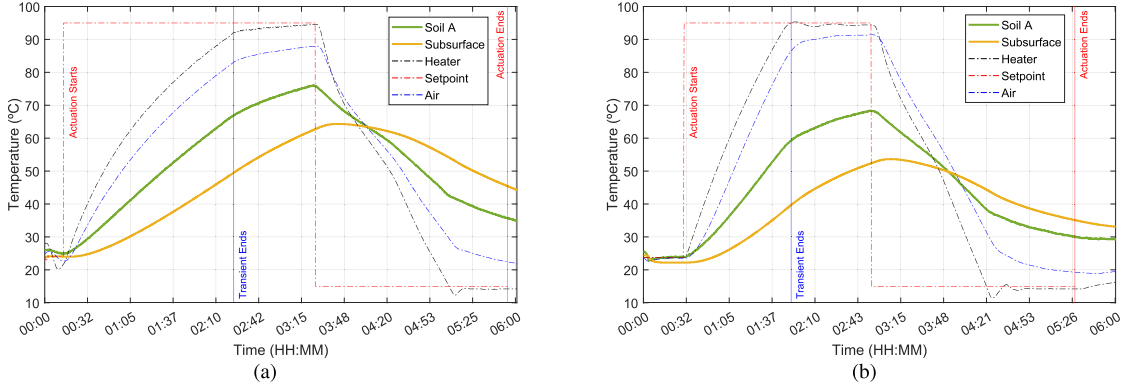


Fig. 11. Diurnal cycle surface and subsurface temperatures of soil A. (a) Earth's pressure (Experiment 1). (b) Martian pressure (Experiment 2).

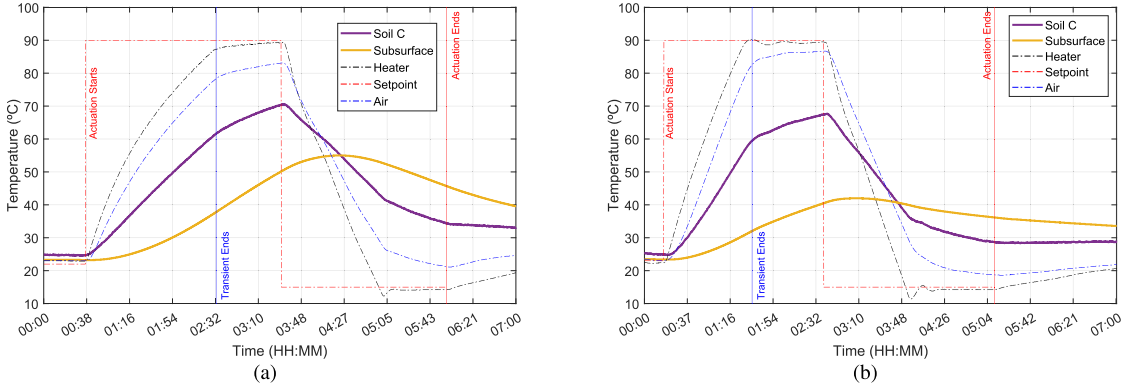


Fig. 12. Diurnal cycle surface and subsurface temperatures of soil C. (a) Earth's pressure (Experiment 3). (b) Martian pressure (Experiment 4).

Second, we compare the values of standard deviation at the end of the transient phase (T_{tran}). Figs. 7(b) and 8(b) offer results for pair-1. At both Earth's and Mars' pressures, T_{tran} renders distinct values for each soil type, which shows that this parameter could be indicative of homogeneity: heterogeneous soils (e.g., soil A) exhibit higher T_{tran} values than homogeneous soils (e.g., soil C). This behavior becomes more pronounced under Martian conditions. A similar analysis can be applied to the graphs of pair-2 [Figs. 9(b) and 10(b)].

Next, we compare the mean surface temperatures with the thermocouples subsurface temperatures of soil A in pair-1 [Fig. 11(a) and (b)]. In this case, the maximum difference between the surface and subsurface temperature is 11.6°C and 14.7°C at Earth's and Mars' pressure, respectively; which

constitutes an increase of 26.72%. Regarding soil C in pair-2 [Fig. 12(a) and (b)], under Earth conditions, the maximum difference is 14.4°C , whereas under Martian-like conditions, it is 24.4°C , constituting a 69.44% increase. In general, the analysis indicates that this difference increases when the pressure decreases, as it gets more difficult for the heat to be transmitted vertically.

Regarding the sinusoidal estimations, I_{sin} in Table III, it is observed that thermal inertia increases when pressure decreases. Soils composed of larger particles, e.g., bedrock, exhibit higher thermal inertia; whereas soils with smaller particles, e.g., soil C, show lower thermal inertia. According to the MEC experimental data, at Earth's pressure, the relative difference between the soil with the highest thermal inertia and

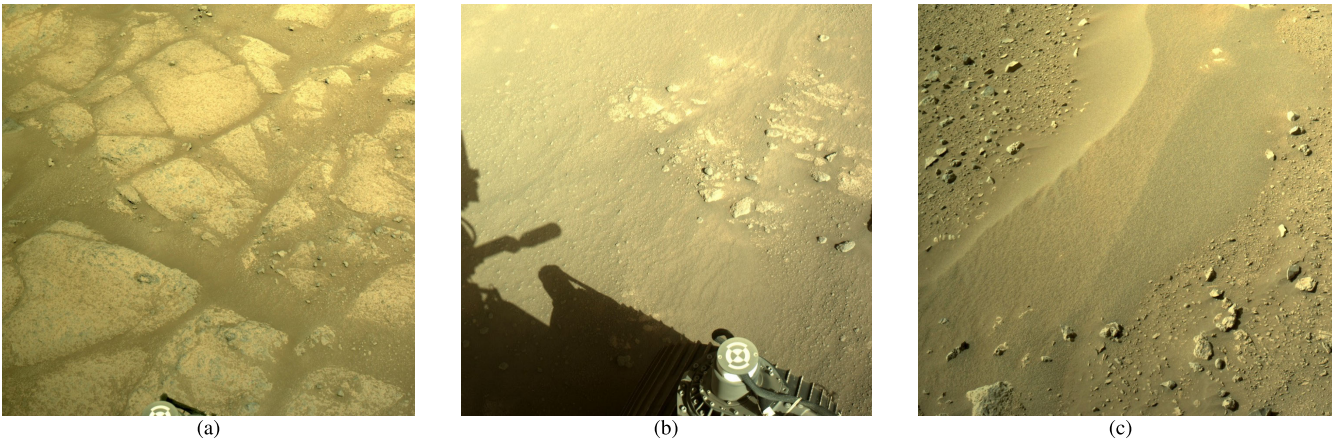


Fig. 13. Examples of Perseverance's Navcam dataset images taken at Sols (Martian days) 193, 237, and 118, respectively. (a) Bedrock. (b) Intermediate soil. (c) Sandy soil.

TABLE III
SURFACE TEMPERATURES AND HEAT FLUXES OF THE SAMPLED SOILS AT EARTH'S AND MARS' PRESSURES AND ESTIMATED THERMAL INERTIA

		T_{tran}	ΔT_s	ΔG_s	I_{sin}
#1 Earth	Bedrock	1.2	53.3	280	309
	Soil A	1.8	51.1	280	322
	Soil B	0.9	51.6	274	312
	Soil C	1.1	51.4	275	314
#2 Mars	Bedrock	1.7	42.0	416	522
	Soil A	2.5	45.7	357	411
	Soil B	0.4	45.5	356	411
	Soil C	1.0	46.3	325	370
#3 Earth	Bedrock	1.3	48.0	268	311
	Soil A	1.8	45.6	265	323
	Soil B	0.8	45.8	264	320
	Soil C	1.1	46.2	257	310
#4 Mars	Bedrock	1.6	38.3	399	548
	Soil A	2.3	41.3	337	431
	Soil B	1.3	41.2	337	431
	Soil C	0.8	43.0	310	379

the one with the lowest thermal inertia is 4.20%. However, at Martian pressure, this difference increases significantly to 42.84%. This observation indicates that soils can be better assessed at Martian pressure compared to Earth's pressure.

C. Comparison With On-Site Mars Thermal Measurements

Even if no actual Martian IR images are available for validation, the consistency of experimental thermal inertia estimations is checked with respect to published analyses of actual rover missions as well as thermopile temperature data from available datasets.

The estimated thermal inertia values under Martian pressure of the bedrock sample bin are consistent with the on-site thermal inertia obtained by Curiosity for bedrock-dominated surfaces (~ 350 – 550 tiu) [26]. Regarding soil C, its estimated thermal inertia closely aligns with surfaces characterized by an average particle size of approximately 1 mm, as indicated by data from Curiosity's observations (~ 265 – 375 tiu) [25]. On the other hand, soils A and B exhibit similar thermal inertia values, despite having different particle sizes. This is consistent with findings indicating that soils with particle sizes

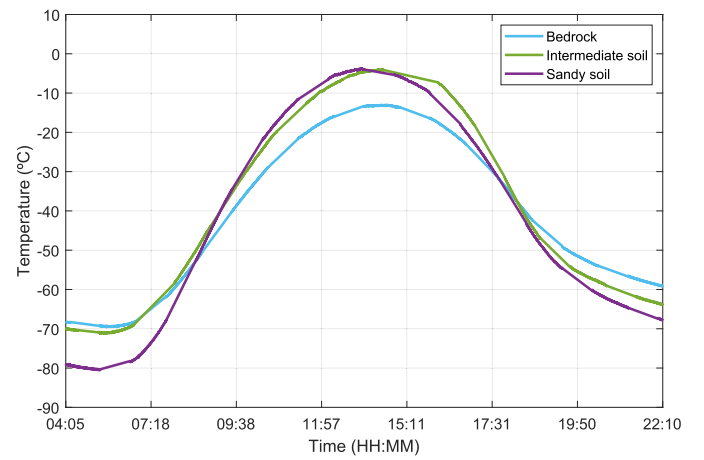


Fig. 14. Diurnal Martian surface temperature recorded by Perseverance's TIRS instrument.

ranging from 1 mm to a few centimeters maintain constant thermal inertia of about 420 tiu [39].

Temperature data from three distinct types of soils, as recorded by the Perseverance rover's TIRS instrument, were extracted from a publicly available dataset¹ provided by NASA. The selection of these soils was based on their visual resemblance to the ones used in our experiments (refer to Fig. 13): Mars' bedrock corresponds to the bedrock sample bin, Mars' intermediate soil is akin to soils A and B, and Mars' sandy soil corresponds to soil C. For the sake of comparison, we selected Perseverance data where the TIRS was pointing at the terrains in the figure for a whole diurnal cycle. The surface temperatures of the three soils during a Martian diurnal cycle are shown in Fig. 14.

In Table IV, we calculated the surface net heat flux values, ΔG_{mars} , as the difference between the maximum and minimum heat flux during that Martian day, using the simplified Martian surface model equation (4) and Mars downwelling SW radiation, R_{sw} , extracted from [12]. Similarly, the increment in surface temperature, ΔT_{mars} , is the difference between the maximum and minimum temperature during that Martian day.

¹<https://pds-geosciences.wustl.edu/missions/mars2020/>

TABLE IV
RELEVANT SURFACE VALUES OF MARTIAN SOILS AND
COMPARISON OF ESTIMATED THERMAL INERTIAS

	R_{sw}	ΔT_{mars}	ΔG_{mars}	I_{mars}	I_{mec}
Bedrock	435	56.6	273	574	535
Intermediate	445	67.2	257	456	421
Sandy	460	76.8	224	348	375

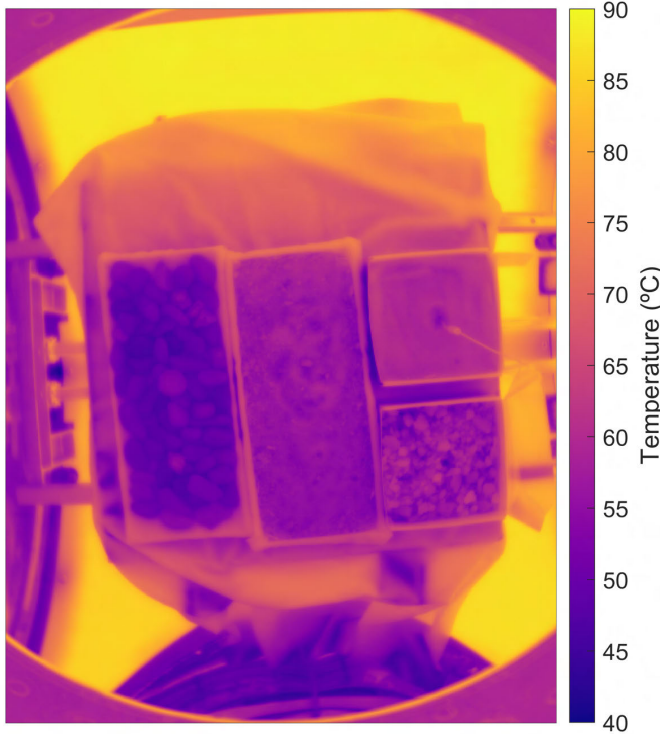


Fig. 15. Example of a thermal image of the sample bins during the experiments.

The thermal inertia estimations based on Perseverance's data, I_{mars} , were calculated using (5). On the other hand, the MEC-based thermal inertia estimations, I_{mec} , were computed as the mean values of the thermal inertia estimations under Mars' pressure of the equivalent soils in Table III.

Similar to the MEC experiments on Martian pressure, three prominent groups can be observed in Table IV based on the surface temperature; from highest to lowest: sandy soil, intermediate soil, and bedrock. Additionally, when comparing the thermal inertia estimations based on Perseverance's data with the MEC-based thermal inertia estimations, a relative error of 6.79%, 7.68%, and 7.76% is observed for the sandy, intermediate, and bedrock soils, respectively.

D. Dataset

During the experiments, we collected a total of 9225 radiometric images, which were saved as plain text 640×480 matrices with each cell containing the temperature in degree Celsius. Snapshots of the thermal images were processed to facilitate direct viewing. An example of one of these snapshots is shown in Fig. 15. Finally, spreadsheets were generated with the heaters, air, and subsurface temperatures recorded by the thermocouples. To the authors' knowledge,

no similar dataset exists in the literature. A public dataset with the recorded data can be found at Zenodo.²

VII. CONCLUSION AND FUTURE WORK

This article has proposed a measurement system configuration and experimental methodology to capture thermal images using MECs to replicate the temperature, pressure, and atmospheric conditions of Martian diurnal cycles. Furthermore, the proposed system has been applied to the UMALASERLAB Mars Environment Chamber to collect a total of 9225 radiometric images and environmental data of four soil types of known physical characteristics for diurnal cycles with Martian and Earth pressures. This unique set of measurements, which has been made publicly available, has allowed us to perform a novel qualitative analysis of soil thermal inertia.

The proposed approach is an innovative solution to address the lack of actual IR images from current Mars rovers, especially in the context of the recent groundbreaking advances in machine learning that demand large amounts of measurements for safe planetary rover navigation based on soil assessment, classification, segmentation, and image understanding. Furthermore, the ability of the system to perform experiments under controlled Earth's and Mars' conditions can contribute to the characterization of future robotic experiments on Earth that aim to replicate planetary conditions accurately.

The possibility to compare the same soil types under controlled Earth and Martian conditions has clearly revealed that the relative difference between thermal inertia extremes (i.e., the distinctiveness of the estimation) of analyzed soils is up to ten times larger in Mars, which corroborates that thermal inertia could offer better soil assessment performance under lower pressures. Furthermore, even if no actual Martian IR images are available for validation, the thermal inertia values obtained in the experiments are consistent with real on-site estimations performed by rovers on Mars.

In summary, the major advantages of the proposed approach with respect to the state-of-the-art methods are as follows.

- 1) High-resolution 2-D thermal imaging allows studying soils with heterogeneous characteristics, which is not possible with current thermopile data from Mars rovers.
- 2) MECs can provide high-fidelity data in contrast to synthetic IR images produced from RGB imagery.
- 3) Experiments can be conducted for soils of known physical characteristics under different controlled conditions (e.g., Earth versus Mars, and diurnal cycles), which is not possible with available thermopile data from Mars rovers, where ground-truth soil characteristics are not available.

Future work may be focused on enhancing the proposed measurement system by using cooled thermal cameras to decrease the minimum temperature that can be remotely measured. Furthermore, the development of control systems capable of producing adaptable sinusoidal temperature profiles in the MEC could help to increase the realism of the physical simulations.

²<http://doi.org/10.5281/zenodo.7750148>

REFERENCES

- [1] J. Y. Wong, *Theory of Ground Vehicles*. Hoboken, NJ, USA: Wiley, 2022.
- [2] R. E. Arvidson et al., "Mars science laboratory curiosity rover megaregion crossings up to sol 710 in Gale Crater," *J. Field Robot.*, vol. 34, no. 3, pp. 495–518, May 2017.
- [3] R. Gonzalez and K. Iagnemma, "Slippage estimation and compensation for planetary exploration rovers. State of the art and future challenges," *J. Field Robot.*, vol. 35, no. 4, pp. 564–577, Jun. 2018.
- [4] L. Feng et al., "An instrumented wheel to measure the Wheel-Terrain interactions of planetary robotic Wheel-on-Limb system on sandy terrains," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [5] R. E. Arvidson et al., "Spirit Mars rover mission: Overview and selected results from the northern home plate winter haven to the side of scamander Crater," *J. Geophys. Res., Planets*, vol. 115, no. E7, pp. 1–19, Jul. 2010.
- [6] D. C. Guastella and G. Muscato, "Learning-based methods of perception and navigation for ground vehicles in unstructured environments: A review," *Sensors*, vol. 21, no. 1, p. 73, Dec. 2020.
- [7] M. G. H. Nampoothiri, B. Vinayakumar, Y. Sunny, and R. Antony, "Recent developments in terrain identification, classification, parameter estimation for the navigation of autonomous robots," *Social Netw. Appl. Sci.*, vol. 3, no. 4, pp. 1–14, Apr. 2021.
- [8] P. Borges et al., "A survey on terrain traversability analysis for autonomous ground vehicles: Methods, sensors, and challenges," *Field Robot.*, vol. 2, no. 1, pp. 1567–1627, Mar. 2022.
- [9] S. Chhaniyara et al., "Terrain trafficability analysis and soil mechanical property identification for planetary rovers: A survey," *J. Terramechanics*, vol. 49, no. 2, pp. 115–128, Apr. 2012.
- [10] N. E. Putzig, *Thermal Inertia and Surface Heterogeneity on Mars*. Boulder, CO, USA: Univ. Colorado at Boulder, 2006.
- [11] G. M. Martínez et al., "Surface energy budget and thermal inertia at Gale crater: Calculations from ground-based measurements," *J. Geophys. Res., Planets*, vol. 119, no. 8, pp. 1822–1838, Aug. 2014.
- [12] G. M. Martínez et al., "Surface energy budget, albedo, and thermal inertia at Jezero crater, Mars, as observed from the Mars 2020 MEDA instrument," *J. Geophys. Res., Planets*, vol. 128, no. 2, pp. 1–23, Feb. 2023.
- [13] C. Cunningham, I. A. Nesnas, and W. L. Whittaker, "Improving slip prediction on Mars using thermal inertia measurements," *Auto. Robots*, vol. 43, no. 2, pp. 503–521, Feb. 2019.
- [14] J. Harrell and M. Largould, "The 25-Ft space simulator at the jet propulsion laboratory," *J. Propuls. Lab.*, Pasadena, CA, USA, Tech. Rep. 32-1415, 1969.
- [15] A. V. Ramachandran, M. I. Nazarious, T. Mathanlal, M.-P. Zorzano, and J. Martín-Torres, "Space environmental chamber for planetary studies," *Sensors*, vol. 20, no. 14, p. 3996, Jul. 2020.
- [16] Z. Wu et al., "A Mars environment chamber coupled with multiple in situ spectral sensors for Mars exploration," *Sensors*, vol. 21, no. 7, p. 2519, Apr. 2021.
- [17] C. Alvarez-Llamas, P. Purohit, J. Moros, P. Lucena, J. Laserna, and J. Vadillo, "A multipurpose thermal vacuum chamber for planetary research compatible with stand-off laser spectroscopies," in *Proc. Lunar Planet. Sci. Conf.*, 2021, p. 2330.
- [18] J. N. Maki et al., "The Mars 2020 engineering cameras and microphone on the perseverance rover: A next-generation imaging system for Mars exploration," *Space Sci. Rev.*, vol. 216, no. 8, pp. 1–48, Dec. 2020.
- [19] H. H. Kieffer, "Thermal model for analysis of Mars infrared mapping," *J. Geophys. Res., Planets*, vol. 118, no. 3, pp. 451–470, Mar. 2013.
- [20] J. C. Price, "Thermal inertia mapping: A new view of the Earth," *J. Geophys. Res.*, vol. 82, no. 18, pp. 2582–2590, Jun. 1977.
- [21] J. Wang, R. L. Bras, G. Sivandran, and R. G. Knox, "A simple method for the estimation of thermal inertia," *Geophys. Res. Lett.*, vol. 37, no. 5, pp. 1–15, Mar. 2010.
- [22] T. N. Carlson, J. K. Dodd, S. G. Benjamin, and J. N. Cooper, "Satellite estimation of the surface energy balance, moisture availability and thermal inertia," *J. Appl. Meteorol.*, vol. 20, no. 1, pp. 67–87, Jan. 1981.
- [23] M. Mellon, "High-resolution thermal inertia mapping from the Mars global surveyor thermal emission spectrometer," *Icarus*, vol. 148, no. 2, pp. 437–455, Dec. 2000.
- [24] J. R. Christian, R. E. Arvidson, J. A. O'Sullivan, A. R. Vasavada, and C. M. Weitz, "CRISM-based high spatial resolution thermal inertia mapping along Curiosity's traverses in Gale Crater," *J. Geophys. Res., Planets*, vol. 127, no. 5, pp. 1–33, May 2022.
- [25] V. E. Hamilton et al., "Observations and preliminary science results from the first 100 sols of MSL rover environmental monitoring station ground temperature sensor measurements at Gale Crater," *J. Geophys. Res., Planets*, vol. 119, no. 4, pp. 745–770, Apr. 2014.
- [26] A. R. Vasavada, S. Piqueux, K. W. Lewis, M. T. Lemmon, and M. D. Smith, "Thermophysical properties along Curiosity's traverse in Gale crater, Mars, derived from the REMS ground temperature sensor," *Icarus*, vol. 284, pp. 372–386, Mar. 2017.
- [27] J. Gómez-Elvira et al., "REMS: The environmental sensor suite for the Mars science laboratory rover," *Space Sci. Rev.*, vol. 170, nos. 1–4, pp. 583–640, Sep. 2012.
- [28] J. Pérez-Izquierdo, E. Sebastián, G. M. Martínez, A. Bravo, M. Ramos, and J. A. R. Manfredi, "The thermal infrared sensor (TIRS) of the Mars environmental dynamics analyzer (MEDA) instrument onboard Mars 2020, a general description and performance analysis," *Measurement*, vol. 122, pp. 432–442, Jul. 2018.
- [29] J. F. Bell et al., "The Mars 2020 perseverance rover mast camera zoom (Mastcam-Z) multispectral, stereoscopic imaging investigation," *Space Sci. Rev.*, vol. 217, no. 1, pp. 1–40, Feb. 2021.
- [30] C. Cunningham, I. Nesnas, and W. L. Whittaker, "Terrain traversability prediction by imaging thermal transients," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2015, pp. 3947–3952.
- [31] R. González, A. López, and K. Iagnemma, "Thermal vision, moisture content, and vegetation in the context of off-road mobile robots," *J. Terramechanics*, vol. 70, pp. 35–48, Apr. 2017.
- [32] Y. Iwashita et al., "Virtual IR sensing for planetary rovers: Improved terrain classification and thermal inertia estimation," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 6302–6309, Oct. 2020.
- [33] D. Atha, R. M. Swan, A. Didier, Z. Hasnain, and M. Ono, "Multi-mission terrain classifier for safe rover navigation and automated science," in *Proc. IEEE Aerosp. Conf. (AERO)*, Mar. 2022, pp. 1–13.
- [34] L. Mandrake et al., "Space applications of a trusted AI framework: Experiences and lessons learned," in *Proc. IEEE Aerosp. Conf. (AERO)*, Mar. 2022, pp. 1–20.
- [35] H. Song, J. Zhang, S. Du, D. Ni, Y. Liu, and Y. Sun, "Constraining the thermal inertia of Mars utilizing machine learning techniques," *Monthly Notices Roy. Astronomical Soc.*, vol. 522, no. 2, pp. 1697–1705, Apr. 2023.
- [36] T. P. Nagle-McNaughton, L. A. Scuderi, and N. Erickson, "Squeezing data from a rock: Machine learning for Martian science," *Geosciences*, vol. 12, no. 6, p. 248, Jun. 2022.
- [37] M. A. Presley and P. R. Christensen, "The effect of bulk density and particle size sorting on the thermal conductivity of particulate materials under Martian atmospheric pressures," *J. Geophys. Res., Planets*, vol. 102, no. E4, pp. 9221–9229, Apr. 1997.
- [38] S. Masamune and J. M. Smith, "Thermal conductivity of beds of spherical particles," *Ind. Eng. Chem. Fundam.*, vol. 2, no. 2, pp. 136–143, May 1963.
- [39] B. M. Jakosky, "On the thermal properties of Martian fines," *Icarus*, vol. 66, no. 1, pp. 117–124, Apr. 1986.
- [40] P. R. Yoder Jr., *Opto-Mechanical Systems Design*. Boca Raton, FL, USA: CRC Press, 2005.
- [41] G. Martínez et al., "The modern near-surface Martian climate: A review of in-situ meteorological data from Viking to Curiosity," *Space Sci. Rev.*, vol. 212, pp. 295–338, Oct. 2017.
- [42] M. Vollmer, *Infrared Thermal Imaging*. Cham, Switzerland: Springer, 2021.



Raúl Castilla-Arquillo (Graduate Student Member, IEEE) received the B.Sc. degree in electronics, robotics, and mechatronics engineering from the University of Seville, Seville, Spain, in 2019, and the M.Sc. degree in mechatronics engineering from the University of Málaga (UMA), Málaga, Spain, in 2020, where he is currently pursuing the Ph.D. degree in mechatronics engineering.

In 2019, he worked for over a year as a firmware engineer in the industry. Afterward, he became a member of the Space Robotics Laboratory, UMA, where he started developing artificial vision algorithms as part of a collaboration agreement between the European Space Agency (ESA) and UMA. His research interests are focused on developing machine learning techniques to improve the perception of space robotic systems using advanced sensors, such as stereo and thermal cameras.



Anthony Mandow (Member, IEEE) received the M.Eng. and Ph.D. degrees in computer systems from the Universidad de Málaga (UMA), Málaga, Spain, in 1992 and 1998, respectively.

He is currently an Associate Professor with the Department of Systems Engineering and Automation, UMA, and a member of the UMA Group for Robotics and Mechatronics. Since 2016, he has been an Academic Coordinator with the UMA Doctoral Program of mechatronics engineering. Since 2021, he has been serving as the Director of the Institute for Mechatronics Engineering and Cyber-Physical Systems, UMA (IMECH.UMA). He has coauthored more than 100 conference and journal papers and has been engaged as a principal investigator in research projects related to field and disaster robotics, multirobot systems, terrain modeling for navigation, artificial intelligence for cyberphysical systems, and innovative practices in higher education.



Carlos J. Pérez-del-Pulgar (Member, IEEE) received the M.S. and Ph.D. degrees in computer science from the University of Málaga (UMA), Málaga, Spain, in 2004 and 2016, respectively.

In 2004, he was given a permanent position as a Research Support Staff with UMA in 2017. Since 2010, he has been a part-time Assistant Lecturer with the Electrical Engineering Faculty, UMA, where he has been responsible for different subjects related to automation and robotics and is currently an Associate Professor and responsible for the Space Robotics Laboratory. He has more than 60 publications on these topics and has been involved in more than 15 different Spanish and international projects. His research interests include machine learning, surgical robotics, space robotics, and autonomous astronomy.



César Álvarez-Llamas received the B.Sc. degree in physics and the M.Sc. degree in analytical chemistry from the University of Oviedo, Oviedo, Spain, in 2011 and 2012, respectively, and the Ph.D. degree (Hons.) in physics from the University of Oviedo, working on the laser-induced breakdown spectroscopy (LIBS) technique in 2017.

As a Post-Doctoral Researcher, he studied portable LIBS equipment with the Carnot Interdisciplinary Laboratory Burgundy (ICB), Dijon, France; the analysis of collimated nanoparticles with Commissariat à l'énergie atomique et aux énergies alternatives (CEA), Saclay, France; and the use of LIBS together with acoustic signals for Mars geo-exploration applications under the NASA Mars 2020 mission in UMALASERLAB, University of Málaga, Málaga, Spain. He is currently working on the applications of kHz LIBS with the Institute Light and Matter (ILM), Lyon, France.



José M. Vadillo received the B.S. degree in biology from the University of Navarra, Pamplona, Spain, in 1989, and the Ph.D. degree in chemistry from the University of Málaga, Málaga, Spain, in 1999, with a dissertation based on the use of pulsed laser combined with optical emission spectroscopy for the analysis of samples in condensed phase.

He has completed scientific stays in different research centers (the Los Alamos National Laboratory, University of Michigan, Ann Arbor, MI, USA; the Swiss Polytechnic Institute, Zürich, Switzerland; and Stanford University, Stanford, CA, USA). He was a Royal Society of Chemistry Scholar in 2000, a Fulbright Scholar from 2000 to 2002, and recruited under the "Ramón y Cajal" Program with the University of Málaga from 2002 to 2007. He was an Assistant Professor with the University of Málaga from 2008 to 2010, where he has been a Permanent Professor since 2010. His scientific interests include instrumental developments and applied spectroscopy, with a spectral emphasis on laser-based methods and mass spectrometry.



Javier Laserna received the degree in chemistry from the University of Granada, Granada, Spain, in 1978, and the Ph.D. degree in analytical chemistry from the University of Málaga, Málaga, Spain, in 1980.

He then joined the faculty of the University of Málaga as an Assistant Professor and became a Full Professor in 1999. He is currently a Co-Investigator for the SuperCam instrument on NASA's Mars 2020 Perseverance rover. He has participated in several research projects funded by the European Commission's Framework Program both as a member of the consortium and as a coordinator. He has been also a principal investigator of more than 25 research projects funded by the Spanish Central Administration and the Regional Administration of Andalucía. He has published more than 300 scientific articles, six books, and book chapters. He is a Co-Inventor of seven patents held by the University of Málaga. He has supervised 35 Ph.D. students. He has delivered numerous invited plenary and keynote conferences at international meetings. His current research interests include the investigation of new measurement principles based on atomic emission, molecular absorption and dispersion, and the understanding of the fundamental phenomena that govern analytical measurements. He is also interested in the development of analytical instrumentation for laser-induced breakdown spectroscopy, remote laser chemical analysis, and online and fieldable analytical measurements. The application areas include the analysis of energetic materials, development of sensors for CBNRE threats, lasers for cultural heritage, and LIBS and Raman spectroscopy for space exploration.

Prof. Laserna was awarded the National Prize for research in analytical chemistry of the Royal Spanish Society of Chemistry in 2009, the National Prize of the Spanish Society of Applied Spectroscopy for his research career in applied spectroscopy in 2010, the 2018 Lester W. Strock Award from the Society of Applied Spectroscopy (USA), and the Ioannes Marcus Marci Medal of the Czech Spectroscopic Society (Czech Republic).